

A DISPROOF OF A UNIFORM LINEAR RAMSEY BOUND FROM A (2, 3)-SPARSITY HYPOTHESIS

ABSTRACT. We show that the proposed estimate

$$R(G, H) \ll m$$

cannot hold uniformly over all graphs G satisfying the condition that every k -vertex subgraph has at most $2k - 3$ edges, even when H has exactly m edges and no isolated vertices. In fact, the obstruction already occurs when G is a path. For every constant $C > 0$ and every positive integer m , there are such graphs G, H with H having m edges and no isolated vertices but

$$R(G, H) > Cm.$$

Thus the proposed statement is false as written. The reason is elementary but decisive: the sparsity condition controls the density of G , not the number of vertices of G , while Ramsey numbers always satisfy a lower bound of order $|V(G)|$ whenever H has an edge.

1. CONVENTIONS AND STATEMENT OF THE PROBLEM

All graphs in this note are finite and simple. For a graph X , write $V(X)$ for its vertex set and $e(X)$ for its number of edges. If X has no isolated vertices and $e(X) = m$, then necessarily $|V(X)| \leq 2m$.

For finite graphs G and H , the Ramsey number $R(G, H)$ is the least positive integer N such that every red-blue edge-colouring of the complete graph K_N contains either a red copy of G or a blue copy of H .

We call a graph G *(2, 3)-sparse* if

$$e(F) \leq 2|V(F)| - 3$$

for every subgraph $F \subseteq G$ with $|V(F)| \geq 2$. The restriction $|V(F)| \geq 2$ is the natural interpretation of the displayed inequality. If one insists on applying the inequality to one-vertex subgraphs, then no nonempty graph satisfies it, since a one-vertex graph has 0 edges but $0 \leq 2 \cdot 1 - 3 = -1$ is false. Under that literal interpretation the question is vacuous. Hence throughout we use the non-vacuous convention $|V(F)| \geq 2$.

The question asks whether the following uniform assertion is true:

There is an absolute constant C such that for every *(2, 3)-sparse* graph G and every graph H with m edges and no isolated vertices, one has

$$R(G, H) \leq Cm.$$

We prove that this assertion is false.

2. TWO ELEMENTARY LEMMAS

We first record two completely standard facts, with proofs included for completeness.

Lemma 1. *Let F be a forest with $k \geq 1$ vertices and $c \geq 1$ connected components. Then*

$$e(F) = k - c.$$

In particular, every forest with $k \geq 1$ vertices has at most $k - 1$ edges.

Proof. Each connected component of a forest is a tree. We prove that a tree on $r \geq 1$ vertices has exactly $r - 1$ edges. This is clear for $r = 1$. For $r \geq 2$, every finite tree has a leaf. Removing a leaf and its unique incident edge gives a tree on $r - 1$ vertices. By induction, that smaller tree has $r - 2$ edges, and therefore the original tree has $r - 1$ edges.

Now let the connected components of F have vertex numbers $r_1, \dots, r_c$. Since each component is a tree, the total number of edges is

$$\sum_{i=1}^c (r_i - 1) = \sum_{i=1}^c r_i - c = k - c.$$

Since $c \geq 1$, this gives $e(F) = k - c \leq k - 1$. $\square$

Lemma 2. *For every integer $n \geq 2$, the path P_n on n vertices is $(2, 3)$ -sparse.*

Proof. Let $F \subseteq P_n$ be any subgraph with $k = |V(F)| \geq 2$. Since P_n is a tree, every subgraph of P_n is a forest. By Lemma 1,

$$e(F) \leq k - 1.$$

For $k \geq 2$ we have

$$k - 1 \leq 2k - 3,$$

because this inequality is equivalent to $k \geq 2$. Therefore

$$e(F) \leq 2k - 3.$$

As F was arbitrary, P_n is $(2, 3)$ -sparse. $\square$

Proposition 3. *Let A and B be finite graphs, and suppose B has at least one edge. Then*

$$R(A, B) \geq |V(A)|.$$

Proof. Let $a = |V(A)|$. Consider the complete graph K_{a-1} and colour every one of its edges red. This colouring has no blue edge at all, hence it contains no blue copy of B , since B has at least one edge. It also contains no red copy of A , because K_{a-1} has only $a - 1$ vertices, while every copy of A uses a distinct vertices. Hence there exists a red-blue colouring of K_{a-1} with neither a red copy of A nor a blue copy of B . By the definition of the Ramsey number, $R(A, B) > a - 1$, and therefore $R(A, B) \geq a$. $\square$

3. THE COUNTEREXAMPLE

Theorem 4. *For every constant $C > 0$ and every positive integer m , there exist finite simple graphs G and H such that:*

- (i) G is $(2, 3)$ -sparse;
- (ii) H has exactly m edges and no isolated vertices;
- (iii) $R(G, H) > Cm$.

Consequently, there is no absolute constant C for which $R(G, H) \leq Cm$ holds for all $(2, 3)$ -sparse graphs G and all graphs H with m edges and no isolated vertices.

Proof. Fix $C > 0$ and a positive integer m . Choose an integer $n \geq 2$ with

$$n > Cm.$$

Let

$$G = P_n,$$

the path on n vertices. By Lemma 2, G is $(2, 3)$ -sparse.

Let

$$H = mK_2,$$

the matching consisting of m pairwise disjoint edges. Then H has exactly m edges. It has no isolated vertices, since every vertex of mK_2 is incident with exactly one edge.

Since H has at least one edge, Proposition 3 gives

$$R(G, H) \geq |V(G)| = n.$$

By our choice of n , this yields

$$R(G, H) > Cm.$$

This proves the asserted counterexample.

If an absolute constant C satisfying $R(G, H) \leq Cm$ for all such G, H existed, the construction above with that same C would produce graphs G, H for which $R(G, H) > Cm$, a contradiction. Hence no such absolute constant exists. $\square$

4. THE FAILURE ALREADY OCCURS FOR ONE EDGE

The preceding theorem uses a matching with an arbitrary prescribed number m of edges. In fact, the obstruction is even simpler: taking $m = 1$ already rules out a uniform bound.

Corollary 5. *For every $n \geq 2$,*

$$R(P_n, K_2) = n.$$

Thus the proposed estimate $R(G, H) \ll m$ fails even when $H = K_2$ and $m = 1$.

Proof. First, Proposition 3 gives

$$R(P_n, K_2) \geq |V(P_n)| = n.$$

For the reverse inequality, consider an arbitrary red-blue colouring of K_n . If at least one edge is blue, then that edge is a blue copy of K_2 . If no edge is blue, then every edge of K_n is red. In that case the red graph is the whole K_n , and K_n contains a Hamilton path on its n vertices, which is a red copy of P_n . Therefore every red-blue colouring of K_n contains either a red P_n or a blue K_2 , so

$$R(P_n, K_2) \leq n.$$

Combining the two inequalities gives $R(P_n, K_2) = n$.

Since n is arbitrary while $m = e(K_2) = 1$ is fixed, no estimate $R(G, H) \leq Cm$ with an absolute constant C can hold uniformly over the class of all $(2, 3)$ -sparse graphs G . $\square$

5. WHY THE PROPOSED HYPOTHESIS IS INSUFFICIENT

The proof above isolates the exact defect in the proposed statement. The condition

$$e(F) \leq 2|V(F)| - 3$$

for every subgraph $F \subseteq G$ controls only the density of G . It gives no upper bound on $|V(G)|$ in terms of $m = e(H)$. But Proposition 3 says that, as soon as H has an edge,

$$R(G, H) \geq |V(G)|.$$

Therefore a linear bound depending only on m is impossible unless some additional hypothesis controls $|V(G)|$ in terms of m or unless the implied constant is allowed to depend on the particular fixed graph G .

This obstruction is not caused by complicated dense graphs. It appears for paths, which are trees, hence acyclic, connected, and much sparser than the displayed $(2, 3)$ -sparsity condition requires.

Remark 6. This note disproves the uniform assertion represented by an unsubscripted estimate $R(G, H) \ll m$, namely an estimate with an absolute implied constant independent of both G and H . A different statement of the form $R(G, H) \ll_G m$, where G is fixed in advance and the implied constant is allowed to depend on G , is not the statement disproved here.

6. VERIFICATION OF THE HYPOTHESES

For completeness, we summarize the checks in the counterexample of Theorem 4.

- (1) The graph $G = P_n$ satisfies the subgraph edge condition: every subgraph of a path is a forest, a forest on k vertices has at most $k - 1$ edges, and $k - 1 \leq 2k - 3$ for all $k \geq 2$.
- (2) The graph $H = mK_2$ has exactly m edges.
- (3) The graph $H = mK_2$ has no isolated vertices, because each of its $2m$ vertices has degree 1.
- (4) The Ramsey lower bound is valid because an all-red colouring of K_{n-1} has no blue edge and therefore no blue copy of any graph with at least one edge, while it has too few vertices to contain a red copy of P_n .

- (5) Since n can be chosen larger than Cm for any prescribed constant C , the ratio $R(G, H)/m$ is unbounded over the allowed pairs (G, H) .

Thus the problem, as stated with a uniform implied constant, has a complete negative answer.